Case 1 — Perfect Competition

A market garden decides what to plant at a price it cannot set.

Micro & Macro Economics · the case, stage by stage

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Scenario figures, check figures and folder paths come from the case brief in the course repository and from the case's stage pages on Kumu. Every stage page carries the full checklist and rubric; see the syllabus for due dates.

REASON

Perfect competition, taught from the supply side — with a spreadsheet a real farmer could use

- A market garden selling into a farmers' market is a price taker: far too small to move the price of a tomato, so the price it faces is simply given — a flat line at any quantity it could plausibly grow.
- That single fact collapses an enormous question, "what should we produce?", into a small one: keep planting while the next bed costs less than it earns, and stop. That rule is P = MC, and it is the whole supply side of the model.
- The capability this case certifies is marginal analysis — read P = MC off a table, off a chart and out of an optimizer's answer; recognize a binding constraint and price what relaxing it is worth; build a constrained optimization a stranger could rerun.

At a price you cannot change, how much should you make?

Source: Shidler perfect-competition-marginal-costs/README.md — the pitch and learning goals; case-perfect-competition.html.

PREDICT

Sixty-four beds, three crops, one season you cannot undo — what do you plant?

- You farm 1.5 acres with one pair of hands and the budget for up to four temporary workers. Nobody will tell you the prices; the farmers' market sets those. The only thing you control is what goes in the ground.
- Every additional bed of a crop makes every bed of it a little more labor-hungry — pest pressure, harvest bottlenecks, walking time — so the cost of the next bed climbs as you plant more.
- Say a mix out loud now, and say why. You will be asked to commit one in writing before any model exists, and you will be asked to defend it afterwards.

You commit the whole season in one decision, and cannot undo it in July.

Source: case-perfect-competition.html — the case lede; Shidler README.md — "the pitch".

REASON

The scenario, in one table — these are the facts, and there are no others

Season 36 weeks · fixed costs $20,000 · 64 beds (16 beds × 4 plots) · the farmer: $50,000 a season, half her time in the field — 720 field hours at an implied $34.72/hr · up to 4 temporary workers at $25,000 each, 1,440 hours each, $17.36/hr

Crop	Max beds	Price $/bed	Labor hrs/wk/bed	Fertilizer $/bed	Diminishing returns
Tomatoes	20	8,800	2.50	880	10.00% / bed
Carrots	20	2,094	0.833	440	2.50% / bed
Mesclun	30	2,700	1.25	880	1.25% / bed

These are the facts your brief restates and your spec encodes.

Source: Shidler README.md — "the scenario — all assumptions in one place"; the same figures on case-perfect-competition-stage2.html.